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RESEARCH ARTICLE

Synthesis of Cu nanoparticles by condensation from the gas phase

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ABSTRACT

A series of computer experiments was conducted by the raw material evaporation and subsequent condensation to determine the most efficient regimes of copper nanoparticles synthesis. By variation of the cooling rate of the metal vapor the formation of Cu clusters were studied. The computer experiments showed the influence of different final temperatures on the shape of the resultant particles. This allowed to determine the conditions for a predominant formation of worm-like clusters of a spherical shape in the gas phase synthesis.

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1. Introduction

Preparing stable nanoscale metal particles is a complicated task, due to their property to rapidly oxidize in air or aqueous environment [1,2]. Nowadays, there are many ways to get nanoparticles of metals, such as mechanochemical methods, plasmachemical methods, some embodiments of chemical, photochemical and radiation reductions and a method of the gas phase condensation [3]. A large number of nanoparticles are also synthesized with high temperature pyrolysis [4,5], electric explosion of wires [6], carbon dioxide [7] and fiber-optical lasers [8].

However, not all the methods can provide the required purity of nanomaterial. From the authors' point of view, the most promising method for the synthesis of nanoparticles as well as for the copper nanoparticles is metals evaporation and subsequent vapor condensation in the inert gas atmosphere. Evaporation of metals can be made in various ways, such as magnetron sputtering [9,10] or powerful electron beam [11,12].

The interest shown by many researchers in nanoscale particles of transition metals, particularly in copper clusters is due to the fact that the size decrease up to several tens of nanometers significantly changes the fundamental properties of the material, e.g. thermal and electrical conductivity, heat capacity and reactivity. Hence, these particles are widely used as a feedstock in the production of various ceramic and composite compounds, superconductors, electronic components, filters, catalysts [1,13,14] and nanoelectronic devices [15]. However, to prepare copper nanoparticles having the required qualities makes high demands on the method production.

This paper considers the most efficient modes of copper nanoparticles synthesis by the gas phase condensation. The disadvantage of an experimental study is impossibility to observe the processes of particles growth and their structuring. The materials obtained by this method are studied only after the synthesis process, without complete understanding of physical processes of nanoparticles formation. Therefore, the experimental study of copper nanoparticles condensation in the gas phase

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[11] was complemented with molecular dynamics (MD) simulation. The computer model was created in the way that it maximally reflected the processes occurring in a real Cu nanoparticles evaporation and condensation experiment.

2. Computer simulation parameters

The computer model of copper nanoparticles condensation after high temperature evaporation of the raw material has been built on the basis of the processes occurring in the experimental–industrial setup. The model system was a cube of 42,600 nm^3 containing 85,000 copper atoms uniformly distributed in space, with initial temperature $T_i = 1000$ K. The meaning of the initial temperature was chosen based on the analysis of the experimental work in metal nanoparticles condensation from the gas phase [16]. Due to the large number of atoms in the system, it was decided to model clusters condensation not in inert gas atmosphere, but in its model. The gas environment in this model consists of virtual particles having a constant temperature while generated by Andersen thermostat. This thermostat allows cooling with fixed speed by means of gradual removing the excess energy from the system. In order to find equations of motion for particles in the model block, the 'speedy' form of the Verlet scheme was used with a constant time step ($h = 2$ fs). The molecular dynamics simulation of copper atoms condensation from the gas phase was made for three different cooling rates $U = 4{\cdot}10^{11}$ K/s, $U = 2{\cdot}10^{11}$ K/s and $U = 0.4{\cdot}10^{11}$ K/s and two final temperatures $T_f = 373$ K and $T_f = 77$ K.

For the purpose of the most correct application of the selected simulation method, it is initially necessary to determine the interaction energy between the atoms and thermodynamic conditions in the system as accurately as possible. A detailed analysis of various articles showed that interaction of atoms during MD simulation synthesis of nanoparticles can be described with a group of many-body potentials. For example, in the article [17], the authors simulated the copper nanoclusters condensation in argon ($N_{Cu} = 5000$ atoms) to analyze the interaction of metal atoms using the potential of embedded atom (EAM-potential), which takes into account pair and many-particle interactions. The authors [18] simulated the atoms condensation of transition metals Ag, Cu, Al and Pt using several interaction potentials. This approach includes the above mentioned EAM-potential and the RGL-potential (Rosato, Guilope, Legrand) developed by Rosato, Guilope and Legrand. Using RGL-potential the authors successfully simulated the condensation process of 1000 Cu and Al atoms in argon with varying pressure.

This paper deals with the modified tight-binding potential within a second-moment approximation (TB-SMA) introduced by Cleri and Rosato [19]. This potential has been successfully applied in a number of studies of nanocrystalline samples [20,21]. Cleri and Rosato potentials are very well established in the modeling of system condensation. For example, the results obtained during condensation of 8000 nickel atoms are well accorded with the experimental data on such basic parameters as particle size distribution, structure and shape [22].

3. Results of computer simulation

During the simulation of the atoms condensation process, the temperature in the system is determined by two independent factors. They are vapor cooling rate and thermal energy produced by the atom bonds energy. The copper atoms condensation in this model runs very rapidly. For the first few picoseconds of condensation, the atoms are combined into small cluster fragments that cause a significant increase of the system kinetic energy, which will undoubtedly entail a sharp leap of temperature. This effect is difficult to observe during direct experimental studies, but this process makes

a significant contribution to the further dynamic growth of particles. The cooling rate of the atomic vapor is determined by the collisions frequency and by the difference between the atomic mass of coolant and of metal atoms. In our case, the gas of virtual particles generated by Andersen thermostat is used as cooling buffer gas.

The processes of the metal atoms free condensation from high temperature gas phase are well known to occur in several stages. The authors of [23] point out three separate condensation stages which may occur practically simultaneously – nucleation, monomeric growth and aggregation or coagulation. The early stage of condensation shows the domination of the spontaneous nucleation of the metal vapor. At this stage, the atoms are combined into small clusters of few atoms – a sort of dimers and trimers interpreted in this work as cluster fragments.

After the formation of several small clusters, the process of particle growth occurs almost immediately, primarily by means of adding one or more monomers or cluster fragments during the modeling step. This process quickly reduces the number of free monomers and brings relatively large clusters into the system. When the number of monomers in the system is significantly reduced, the particle growth is mainly regulated by cluster aggregation process [24].

Further, these stages of high temperature nucleation process are considered in a more detailed way to take into account the system transition to a low temperature phase of aggregation. Figure 1 shows typical time dependence of the number of clusters in the system and 'snapshots' of condensation for the first 0.35 ns of simulation. In the process of growth for the simulation time $t = 0.1$ ns, the copper atoms form quite large liquid clusters of several hundred atoms size. And since the temperature in the system is still rather high, the atoms can freely leave the surface of the cluster, which leads to uneven particles growth (Figure 1(b)). But the gradual removal of heat out of the system speeds up the nucleation process so that after 10 picoseconds it brings the formation of massive amorphous-like cluster aggregations of an irregular external shape.

This condensation stage is evidently shown in Figure 1(c). Therefore, it is a turning point as the biggest number of large particles has already been formed, but they are still of a molten state. However, after a few picoseconds of simulation, there begin irreversible processes of nanoclusters crystallization and structuring. They combine with each other by means of agglomeration and aggregation, and form further final particles.

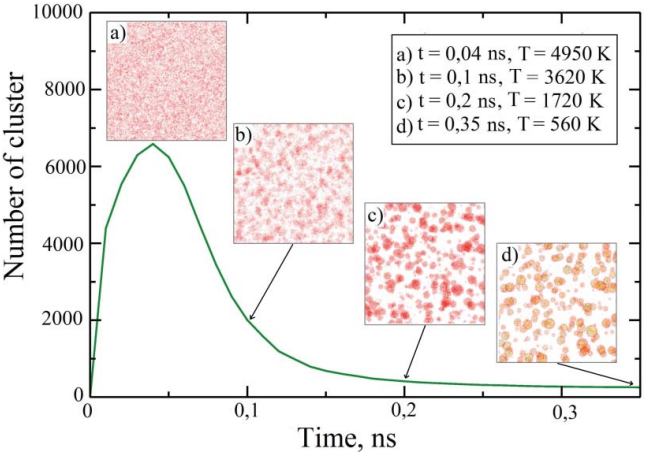


Figure 1. Time dependence of the quantity of clusters in the 85,000 copper atoms system with the cooling rate $U = 2 \cdot 10^{11}$ K/s and final temperature $T_f = 77$ K. (a)–(d) – 'Instantaneous snapshots' of the system.

Table 1. The structure and shape of Cu nanoclusters synthesized from the gas phase.

Cooling rate (U, 10^{11} K/s)	Final temperature (T_f, K)	Number of clusters	Average cluster size	Structure				Form	
				FCC (HCP) (%)	Ih (%)	Dh (%)	Amorphous (%)	Worm-like (%)	Spherical (%)
4	77	46	1847	22.02	45.83	20.87	11.28	76.08	23.92
2	77	35	2428	22.51	41.93	30.02	5.54	74.28	0.72
0.4	77	17	5000	23.59	26.82	49.59	0	58.83	41.17
4	373	19	4473	22.57	31.38	46.05	0	89.47	10.53
2	373	13	6538	29.43	30.07	40.5	0	84.61	15.39
0.4	373	6	14166	38.33	26.67	35	0	83.34	16.66

Note: Dh, decahedral; FCC, face centered cubic; HCP, hexagonal close packed; Ih, icosahedron.

For the simulation time $t = 0.35$ ns (Figure 1(d)), the clusters crystallization almost comes to the end, but the size of these particles is still small; the largest cluster contains only 1900 atoms. This step completes the first and the most important condensation stage, namely the stage of nucleation, growth and crystallization of clusters. At the following stage, the spontaneous association of crystalline particles functions as fairly large agglomerates.

Six modeling options of copper atoms condensation from the gas phase were held under three different values of the cooling rate and two final temperatures. This resulted in a detailed analysis of the simulated systems at the last stage of the synthesis. Particularly, there studied the effect of variable parameters on the amount, structure and shape of the obtained particles (Table 1). It is clearly seen that the rate of the gas mixture cooling directly affects the number and average size of clusters in the system providing the same number of atoms (85,000).

The cooling rate reduction by 10 times causes the increase of the particles average size by 2.7 times with final temperature of 77 K and by 3.1 times with $T_f = 373$ K. Besides, in the system cooled up to liquid nitrogen temperature, the number of particles is approximately two and a half times more than that of the ones in the system cooled up to the boiling point of water (Table 1, Figure S1). This dependence is quite natural, and as previously mentioned, it is associated with an excess of kinetic energy in the cooling water system. The size of the particles during the computer experiments varied from few to tens of nanometers [25], and the particles synthesized under such conditions could be formed by combining each other into more than 20 small primary clusters (Figure 2, Figure S2). The process of experimental synthesis of copper particles forms clusters of a similar worm-like shape, but of a much larger size (Figure 3(b)). The formation of large particles during the experiment is explained due to the fact that cooling and formation of nanoparticles take much more time than their simulation does.

The color of the atoms (Figure 2) shows the result of Common Neighbour Analysis analysis in the following way – atoms drawn blue (green) have local crystalline face centered cubic, FCC (HCP, hexagonal close packed) environment; it implies that the number of neighbors for these atoms in the first coordination sphere is $Z = 12$; the number of neighbors in the second sphere equals to 6 and respectively in the third sphere it makes up 24 (FCC) or 2 (HCP). The yellow atoms have no identified structure. However, their coordination number is $Z = 12$, that is the number of neighbors in the second coordination sphere is <6. The atoms with coordination numbers $Z < 12$ are drawn red.

This work also calculates the surface area of Cu worm-like nanoparticles synthesized from the gas phase, being sized from 648 to 11,770 atoms (Figure 4). The comparison of the surface area for worm-like Cu clusters and spherical nanoparticles cut from FCC lattice with the same amount of atoms was produced. Therefore, when the atoms increase, the surface area of worm-like nanoparticles grows ~1.5 times quicker than the surface area of spherical nanoparticles. It means that worm-like Cu nanoparticles may be more effectively used in various chemical reactions in contrast to spherical nanoparticles due to the worm-like nanoparticles increase.

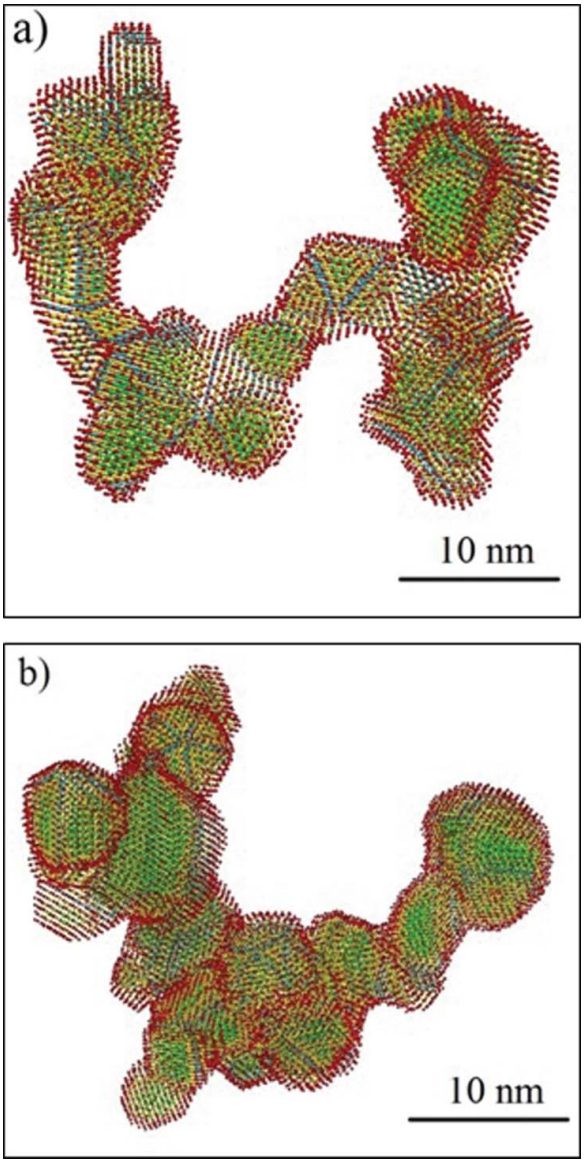


Figure 2. Examples of copper nanoclusters synthesized from the gas phase (a) $N = 17{,}916$ atoms, (b) $N = 44{,}270$ atoms.

4. Conclusions

The investigation of copper nanoparticles production by the source material evaporation and following condensation was held to determine the best mode of synthesis of the material with the desired physical properties. The molecular dynamics simulation of copper nanoparticles condensation was carried out at different cooling rates and final temperatures.

The result of the computer simulation is that the gas mixture cooling rate and the final temperature directly affect the number and size of the particles of the system providing the same number of atoms. Since the cooling rate decreases by 10 times, the average size of the obtained particles increases by 2.7 times at final temperature of 77 K and by 3.1 times at $T_f = 373$ K.

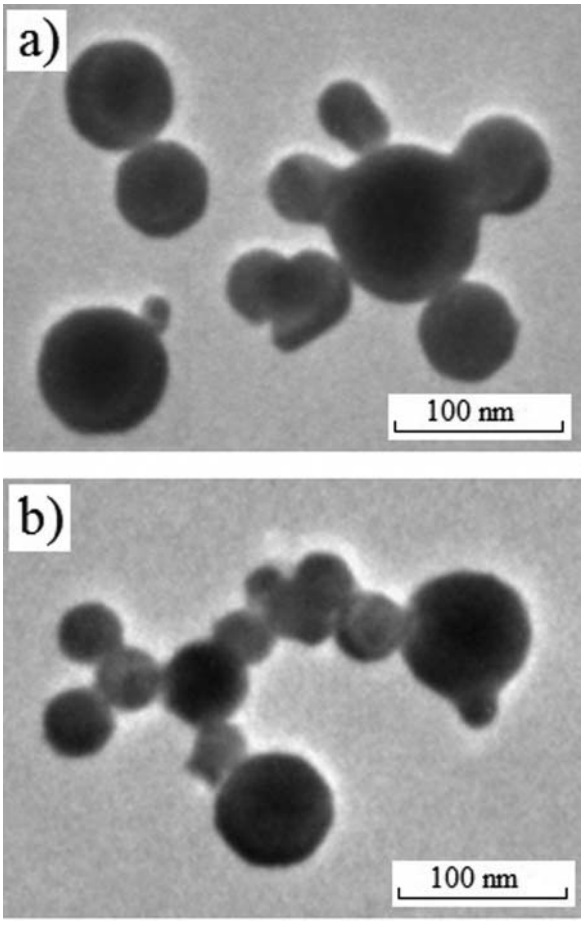


Figure 3. Transmission electron microscopy (TEM) image of copper nanoparticles condensed from the gas phase.

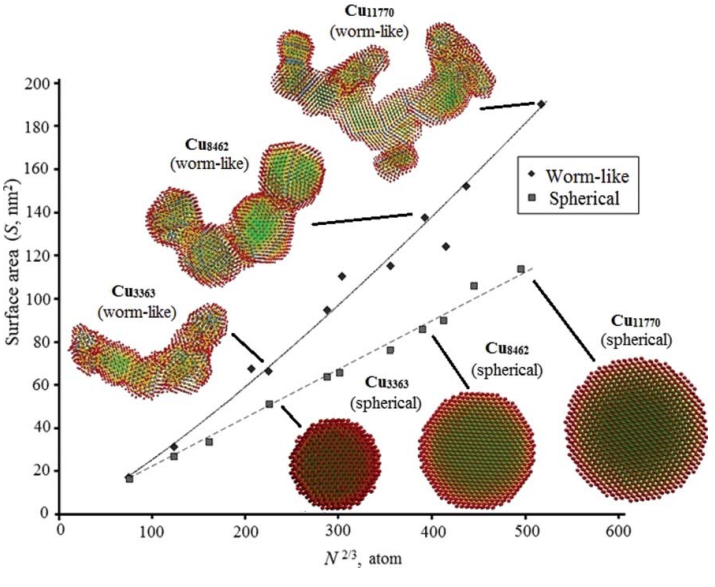


Figure 4. The dependence of the surface area (S, nm^2) on cluster sizes ($N^{2/3}$, atoms).

The computer experiment investigated the influence of different final temperatures on the shape of the resultant particles. As a result, the authors determined the conditions for the predominant formation of the worm-like clusters of a spherical shape during the gas phase synthesis. Also, it was determined that the surface area of worm-like nanoparticles grows ~1.5 times quicker than the surface area of spherical nanoparticles. The spherical clusters production can be performed most efficiently in a system with the coolant temperature of about 77 K, while worm-like clusters are formed at the higher cooling temperature ($T_f = 373$ K).

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Disclosure statement

No potential conflict of interest was reported by the authors.

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